Agenda

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- What is a Database Management System (DBMS)?
- The Three-Schema Architecture: External, Conceptual, Internal
- Entity-Relationship (ER) Model: entities, attributes, relationships
- From ER diagram to relational schema
- Why data modelling matters for e-business applications

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Relational Model Essentials

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- A relation is a table; a tuple is a row; an attribute is a column
- Candidate key vs. primary key vs. foreign key
- Referential integrity protects relationships between tables
- SQL families: DDL (CREATE), DML (SELECT/INSERT/UPDATE), DCL (GRANT)
- Indexes trade storage for faster query response time

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Normalization

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- 1NF: atomic values, no repeating groups
- 2NF: remove partial dependency on part of a composite key
- 3NF: remove transitive dependency (non-key depends on non-key)
- BCNF: every determinant must be a candidate key
- Goal: eliminate insertion, update and deletion anomalies

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Transactions & ACID

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- A transaction is a logical unit of work
- Atomicity: all or nothing
- Consistency: constraints hold before and after
- Isolation: concurrent transactions do not interfere
- Durability: committed changes survive system failure

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